

This is a conversation I have had with chat gpt where we brainstormed everything get the idea of what we are building from this

So i want to build a pos system for a restaurant called glitos. Now i know exactly how they operate where when a customer comes in they write their irder in a receipt then the one in the couter passes it to the server who passes the food to the waiter. At the end of the day they tally the days sales in a book to know the total amount of money sold. So i want to build it and i want it to be pwa system. Ask me questions about the business to help me plan and build it

Since you already understand how they work, don't start by designing screens. Start by mapping the business. A POS that matches their workflow will be much easier for them to adopt than forcing them to change how they work.

I'll ask the questions in phases.

1. Restaurant Basics

- Is it dine-in only, takeaway only, or both?
 - Do they do deliveries?
 - How many branches are there?
 - About how many customers do they serve per day?
 - How many employees use the system?
-

2. Staff Roles

Who exactly works there?

Example:

- Cashier
- Waiter
- Kitchen staff
- Manager
- Owner

For each role:

- What are they allowed to see?
 - What are they allowed to edit?
-

3. Current Ordering Process

Walk me through one customer.

Example:

Customer enters →

Cashier writes receipt →

Customer pays? or pays later? →

Receipt goes to kitchen →
Kitchen prepares food →
Waiter serves →
Order completed.

Questions:

- Does the customer pay before food is prepared or after eating?
 - Does the cashier keep one copy and kitchen another?
 - Are receipt books numbered?
 - Can an order be changed after it reaches the kitchen?
 - Can an order be cancelled?
-

4. Menu

- How many food items are there?
 - Do prices change often?
 - Can one order contain multiple items?
 - Are there sizes? (Small, Medium, Large)
 - Are there add-ons? (Extra cheese, extra chips)
-

5. Payments

What payment methods do they accept?

- Cash
- M-Pesa
- Card
- Mixed payment

Questions:

- Do they print receipts?
 - Is an M-Pesa message enough?
 - Can customers pay partly in cash and partly by M-Pesa?
-

6. Kitchen Workflow

When the kitchen receives an order:

- Is food prepared in order?
 - Is priority ever changed?
 - Can multiple cooks prepare one order?
 - How do they know an order is ready?
-

7. Waiters

- Do waiters have assigned tables?
 - Can one waiter serve multiple tables?
 - Can multiple waiters work on one order?
-

8. Tables

- How many tables?
 - Are table numbers used?
 - Can customers move tables?
 - Can two tables be combined?
-

9. End of Day

Currently they total sales in a book.

Questions:

What else do they calculate?

- Cash received
 - M-Pesa received
 - Expenses
 - Profit
 - Number of customers
 - Best-selling item
-

10. Inventory

This is a big one.

Do they currently track:

- Chicken pieces
- Chips
- Soda
- Cooking oil
- Flour
- Packaging

Or do they only count sales?

11. Owner Reports

If the owner opens the app tonight, what should they immediately see?

Example:

- Today's sales
 - Orders today
 - Cash vs M-Pesa
 - Best-selling food
 - Busy hours
 - Staff performance
 - Low stock
-

12. Devices

Since you're making a PWA:

- Will everyone use one computer?
 - Will the cashier have a laptop?
 - Will the kitchen have a tablet?
 - Will waiters have phones?
 - Will everything run over local Wi-Fi?
-

13. Internet

- Does the restaurant always have internet?
 - Should the POS continue working offline?
 - When internet returns, should it sync automatically?
-

14. Future Features

Which of these do you eventually want?

- Customer loyalty points
- QR code menu
- Online ordering
- Reservations
- Kitchen display screen
- SMS notifications
- M-Pesa STK Push integration
- Expense tracking
- Payroll
- Inventory management
- Multiple branches

- Owner dashboard from home
-

15. Technical Questions

Since you'll build it yourself:

- Frontend: HTML/CSS/JS or React?
 - Backend: Node.js/Express?
 - Database: SQLite, PostgreSQL, or something else?
 - Authentication: username/password or PIN?
 - Will it be hosted online or run on a local server inside the restaurant?
-

One suggestion before you write any code

Think of the POS as **five connected apps** rather than one:

1. **Cashier App** – creates orders, takes payments, prints receipts.
2. **Kitchen Display** – shows incoming orders, marks them as preparing and ready.
3. **Waiter App** – sees ready orders and delivers them.
4. **Manager Dashboard** – sales, reports, inventory, staff.
5. **Owner Dashboard** – view everything remotely from any device.

If you build it this way, you can start with just the cashier app and gradually add the others without rewriting the system.

My first question is this: **Describe exactly what happens from the moment a customer walks into Glitos until they leave. Include every step, even small ones like who writes the receipt, who keeps it, when payment happens, and who hands the food to the customer.**

They do both. Yes. One. A lot. 4. so i want the organization they already have to stay but they should use receipts from a small thermo printer. Customers pay after they have eaten also some are given credit. Like 20 food items. An order can contain multiple items. No size difference like large etc no addons. Cash and mpesa. I want an stk push and they can pay partly in cash and mpesa. No assigned sits for waiters food is already made so just serving and packing. Since they do delivery i want there to be a booking or ordering app for the customers who want delivery. At the end of the day they calculate everything. Do an inventory trucking. The owner should be able to see all those things. They will use their phones run over wifi. Always have internet but when it is down it will save locally and update when online. Will use mern i will have all the 5 connected apps

That's a smart decision. **Don't build KRA eTIMS into version 1.** Get the restaurant using the system first, then add eTIMS once the core POS is stable.

I would structure the project in phases.

Phase 1 – Core POS (MVP)

- User login and roles
- Menu management
- Create and edit orders
- Thermal receipt printing
- Kitchen/serving workflow
- Cash, M-Pesa, mixed payments, and credit
- Inventory deduction
- End-of-day reports
- Offline support with automatic sync
- Owner dashboard

This alone will replace their paper workflow.

Phase 2 – Customer Experience

- Customer delivery app/PWA
- Order tracking
- STK Push payments
- WhatsApp order confirmations
- Delivery management

Phase 3 – Business Intelligence

- Advanced inventory
- Supplier management
- Expense tracking
- Profit analysis
- Customer loyalty
- Analytics and trends

Phase 4 – KRA eTIMS

At this stage, every completed sale can be submitted to eTIMS.

Design your backend now so adding eTIMS later doesn't require rewriting everything. A good approach is to have a separate **Tax Service** or **Fiscal Service** in your backend.

For example:

```
Order Completed
  ↓
Payment Recorded
  ↓
Receipt Generated
  ↓
Tax Service
  ↓
eTIMS (enabled later)
```

Initially, the Tax Service can simply do nothing after generating the receipt. Later, you connect it to the KRA API. That way, the rest of your application doesn't need to change.

One more suggestion: create an **audit log** from the beginning. Every important action should be recorded, such as:

- Who created an order
- Who edited it
- Who cancelled it
- Who accepted a payment
- Who changed inventory
- When a receipt was reprinted

This becomes invaluable if there's ever a dispute about sales, stock, or cash shortages.

From what you've described, this isn't just a POS anymore—it's shaping up to be a full restaurant management system that can later support multiple restaurants with configurable workflows. If you design the data model carefully now, Glitos can be your first deployment and your test case before expanding to other businesses.

This is what we are building